Trust at Scale: The Economic Limits of Cryptocurrencies and Blockchains

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Overview

1. Introduction
2. Nakamoto Trust: A Critique in 3 Equations
3. Comparison of Nakamoto Trust and Traditional Trust
4. Conclusion

1 Introduction

2 Nakamoto Trust: A Critique in 3 Equations

3 Comparison of Nakamoto Trust and Traditional Trust

4 Conclusion

Government and Laws

- Economists have long agreed that the market system requires some form of government and rule of law for support.
  1. Smith (1776): "commerce and manufactures can seldom flourish long in any state" without a legal system, property rights and contract enforcement.
  2. Hayek (1960): to maximize freedom, defined as absence of coercion, it is necessary to have a government with the power to coerce.
  3. Friedman (1962): government sets "rules of the game" and serves as "umpire".

Nakamoto's Invention

- Satoshi Nakamoto (2008) proposed a new economic system without using legal trust.
- Trust arises from cryptography and economic incentives.
  - "Permissionless consensus"
  - Completely anonymous and decentralized
  - Large, freely-entering and -exiting set of participants collectively maintains a common data set.
  - No trusted parties
- Cryptocurrencies including Bitcoin

Main Results

This paper studies the economic limitations of Nakamoto's form of trust.

Specifically, this paper

- Establishes 3 equations to show that Nakamoto trust is unreasonably expensive.
  1. Eq. (1): zero-profits condition;
  2. Eq. (2): incentive compatibility condition;
  3. Eq. (3): (1) + (2) $\Rightarrow$ very large "flow" cost of honest trust support.
- Emphasizes the key difference between Nakamoto trust and traditional trust: economies of scale.

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Setup

- I: number of honest participants
  - Each player i chooses a quantity $x_i \in \mathbb{R}_+$ of "trust support".
  - $N = \sum_{i=1}^{I} x_i$: total quantity of trust support
- c: cost per unit time to supply one unit of trust support
- p_{block}: economic reward paid to a player who adds a new block of transactions
  - Player i wins p_{block} with prob. $\frac{x_i}{N}$.

Honest Miners' Profits

- Assume that one block is solved per unit time.
- Profits per unit of trust support per unit time are

$$\frac{1}{N} p_{block} - c.$$

  - N units of trust support solves a block per unit time.
  - Each of the N units is equally likely to win.

Zero-Profit Condition (Honest Play)

- Q1: how much "trust support" will maintain the permissionless consensus if we restrict all to behave honestly?

Definition 1

A zero-profit honest Nakamoto trust equilibrium[a] *consists of quantities* $\{x_i^*\}_{i=1}^{I}$ *and a difficulty level* D^* *such that participants (i) solve one block per unit time (as a normalization), and (ii) earn zero economic profits in expectation.*

[a] A special case of Nash equilibrium

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Zero-Profit Condition (Honest Play)

Proposition 1

Let $N^* = \sum_{i=1}^{I} x_i^*$. In any zero-profit honest Nakamoto trust equilibrium,

$$N^* c = p_{block}. \tag{1}$$

PROOF

- $\frac{1}{N^*} p_{block} - c = 0 \Longrightarrow N^* c = p_{block}.$

Incentive Compatibility Condition (Majority Attack)

- One dishonest player
  - Attack by choosing AN^* units of trust support, $A > 1$
  - Obtain an $\frac{A}{A+1}$ majority of trust support
  - At cost AN^*c per unit time
- $t(A)$: expected duration of the attack
  - $AN^*ct(A) = N^*c \cdot At(A)$: gross cost of the attack
  - He chooses A^* to minimize $A \cdot t(A)$.
- V_{attack}: value of attack.

Incentive Compatibility Condition (Majority Attack)

- Q2: how much security is generated by the amount of honest trust support in (1)?
  - Majority attack: weakness of permissionless consensus

Definition 2

Nakamoto trust is incentive compatible (IC) *against an outsider attack, on a gross-cost basis, if the gross cost of attack exceeds the benefits of attack:*

$$A^* N^* ct(A^*) > V_{attack}. \tag{2}$$

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Equilibrium Constraint

The Problem

$$N^* c = p_{block} \tag{1}$$

$$A^* N^* c t(A^*) > V_{attack} \tag{2}$$

Theorem 1

The zero-profit honest Nakamoto trust equilibrium condition (1) and the IC against outsider attack condition (2) together imply the equilibrium constraint:

$$p_{block} > \frac{V_{attack}}{A^* \cdot t(A^*)}. \tag{3}$$

PROOF

- $p_{block} = N^* c > \dfrac{V_{attack}}{A^* \cdot t(A^*)}.$

Equilibrium Constraint

$$p_{block} > \frac{V_{attack}}{A^* \cdot t(A^*)}. \tag{3}$$

- Very expensive form of trust!
  - Linear in amount of V_{attack}
    - e.g. a $1B attack is 1000x more expensive to prevent than a $1M attack.
  - Imagine a brand only as trustworthy as its flow investment in advertising.
  - Imagine a country only as secure as its flow expenditure on soldiers at the border.

Thought Experiment

- Consider "double-spending" attack.
- Thought experiment: what is the equilibrium cost of security for a given dollar value of V_{attack}?
  - Even 1-week escrow period requires an annual expense of $48B to keep Bitcoin secure up to $1B attack.
  - 5% of Global GDP to secure against $100B attack

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Traditional Trust

- A transaction between two parties of size V. One party has an opportunity to cheat and steal the other party's assets.
  - Party 1 chooses an action from {Engage, Don't Engage}.
  - Party 2 chooses an action from {Honest, Cheat}.
- Payoffs:
  1. Engage + Honest: both get $\tau > 0$ as the net benefit of an honest transaction.
  2. Engage + Cheat: Party 2 gets V and Party 1 gets $-V$.
  3. Don't Engage: both get 0.

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Nash Equilibrium

- For simplicity, set $\tau = 5$ and $V = 100$.
- Nash equilibrium: given the other player's strategy, one chooses his best strategy as a response.

	Honest	Cheat
Engage	$\underline{(5,5)}$	$\overline{(-100,100)}$
Don't Engage	$\overline{(0,0)}$	$\underline{\overline{(0,0)}}$

- (Don't Engage, Cheat) is the only Nash equilibrium.

Legal System

- Add a legal system with the power to enforce contracts.
- If Party 2 plays Cheat, Party 1 can pay c_l to bring the matter to court.
- The court can perfectly observe whether Party 2 played Cheat.
  - If Party 2 played Cheat, it can compel Party 2 to return Party 1's assets and punish Party 2 with a large fine f.
  - Party 1's payoff is $-c_l$.
  - Party 2's payoff is $-f$.
- Set $c_l = 10$, $f = 100$.

Nash Equilibrium with a Legal System

	Honest	Cheat
Engage	$\underline{\overline{(5,5)}}$	$(-10,-100)$
Don't Engage	$\overline{(0,0)}$	$\underline{\overline{(0,0)}}$

- (Engage, Honest) is a Nash equilibrium.
- The players will transact honestly on the equilibrium path.
- The court need not involve itself with most transactions!
  - Deterrence effect

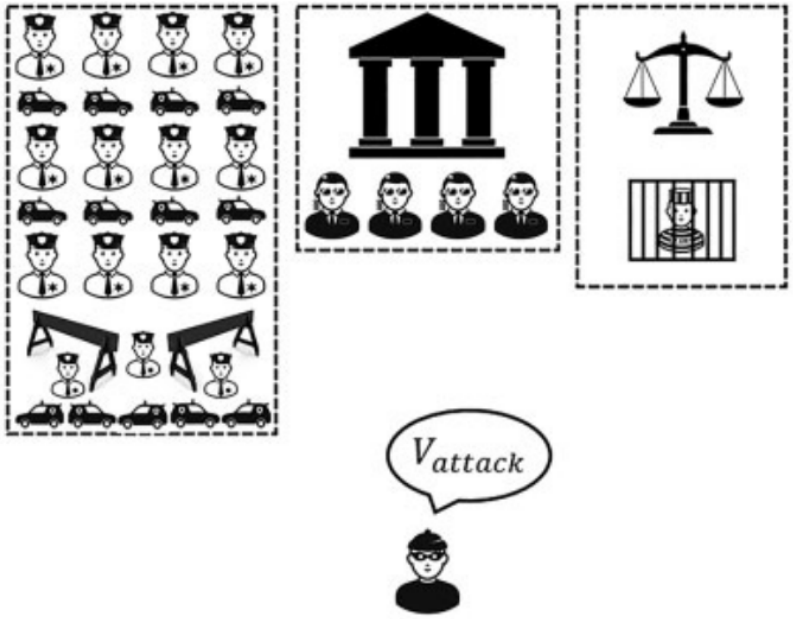
Nakamoto Trust v.s. Tradional Trust

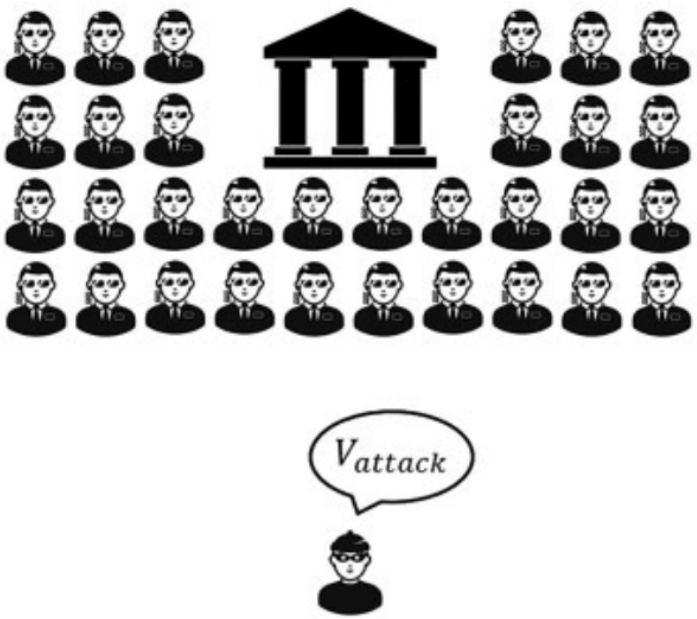
- V_{honest}: average volume transacted per period if trust is secured.
- Traditional trust costs
  - a fixed cost F, and
  - a variable cost per unit transacted of c.
  - Society's cost of traditional trust per period is $F + cV_{honest}$.
- The two trust models' cost per unit volume are:

$$\text{Traditional Trust:} \quad \frac{F}{V_{honest}} + c$$

$$\text{Nakamoto Trust:} \quad \frac{V_{attack}}{V_{honest}}$$

Illustration





Traditional Trust	Nakamoto Trust
cost of overcoming guards + cost of overcoming police reinforcements + risk × punishment if caught $> V_{attack}$	cost of overcoming guards $> V_{attack}$

Figure 1: Nakamoto Trust v.s. Traditional Trust

Economy of Scale

$$\text{Traditional Trust:} \quad \frac{F}{V_{honest}} + c$$

$$\text{Nakamoto Trust:} \quad \frac{V_{attack}}{V_{honest}}$$

- Economy of scale for traditional trust
  - A society that pays a fixed cost of operating a court system can facilitate honest transactions.
    - Zero marginal cost of security due to the deterrence effect.

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Comparison of Trust Models

	Cost to Break Trust	Variable Cost of Honest Play	Fixed Cost Investments that Support Honest Play
Nakamoto Trust in its original form (Natamoto, 2008)	Flow. Attacker pays cost of honest trust support $p_{block} = N^* c$ for a short period of time.	p_{block} per period.	–
Repeated Interaction (Schelling, 1956; Aumann, 1959)	Stock. Attacker loses the net present value of the trust relationship $V_{trust} = \frac{1}{1-\delta}\tau$.	0, given pre-existing trust relationship.	Investment in trust relationship, brand, reputation.
Credible Deterrence (Hayek, 1960; Becker, 1968)	Arbitrarily high up to value of life and assets. Attacker can be punished by the state.	"Security guards" in Figure 1. Variable costs of monitoring and enforcement associated with specific locations or transactions.	Investment in government and legal system.
Collateral	Stock + Punishment. Attacker loses the value of their collateral plus any additional punishment.	0, under conditions of Modigliani-Miller theorem. Requires legal enforcement mechanism which entails fixed costs.	Collateral value.

Figure 2: Comparison of Trust Models

Conclusion

- Anonymous, decentralized trust enabled by Nakamoto (2008) blockchain: *ingenious but expensive*
- Permissionless consensus has been of limited usefulness in its pure form.
  - The economic limits suggest this is intrinsic to the technology.
- This paper also sharpens our understanding of what is special about traditional forms of trust: economies of scale.
- It is exactly the aspect of Nakamoto (2008) that is so innovative relative to traditional distributed databases – *the anonymous, decentralized trust* – that is the source of its economic limits.

More

This presentation omits the following topics:

- Detailed analysis of double-spending attacks
- Discussion on the case where an attacker loses their capital
- A large strand of CS literature on Nakamoto trust